DATE: 12.09.2025

RISK ANALYTICS REPORT 1

1. Data Preparation

- Loaded **Trading (30k rows, 20 cols)** and **Instruments (1.5k rows, 18 cols)** datasets.
- Merged into a unified dataset with **38 features**.
- Verified data integrity with **SHA-256 snapshots** for reproducibility.

2. Portfolio & Strategy Performance

- Computed **daily portfolio P&L** and **strategy-level P&L**.
- Normalized results into daily returns for consistent analysis.

3. Risk Metrics

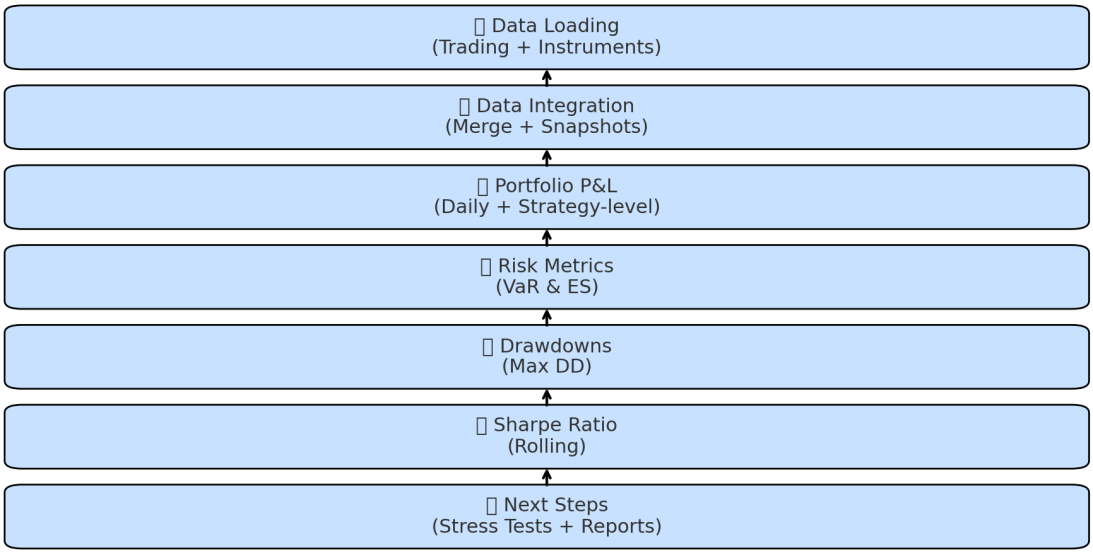
- Calculated **Value at Risk (VaR)** and **Expected Shortfall (ES)** at 95% and 99% confidence levels.
- Produced **strategy-wise risk metrics** for deeper comparison.

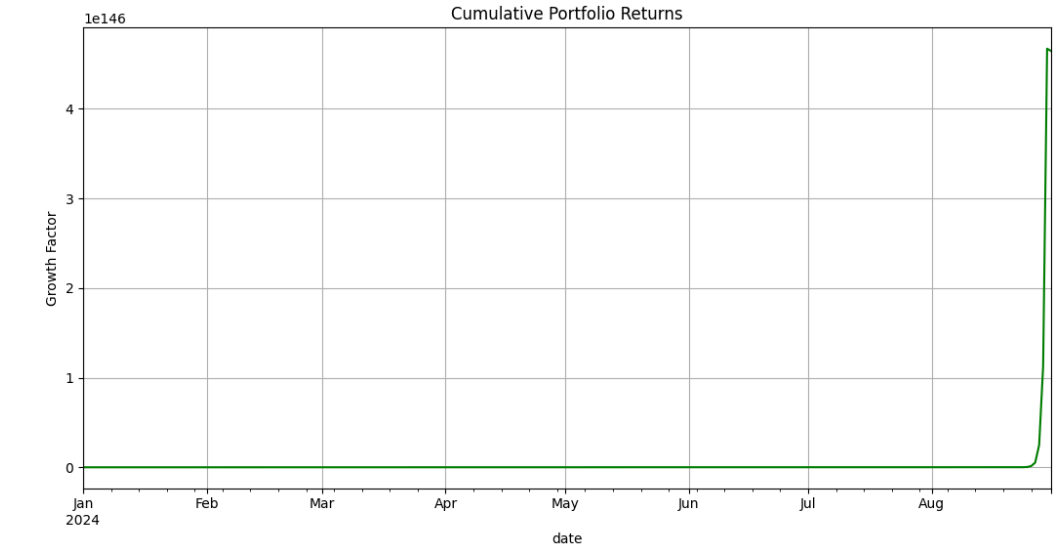
4. Drawdowns & Sharpe Ratios

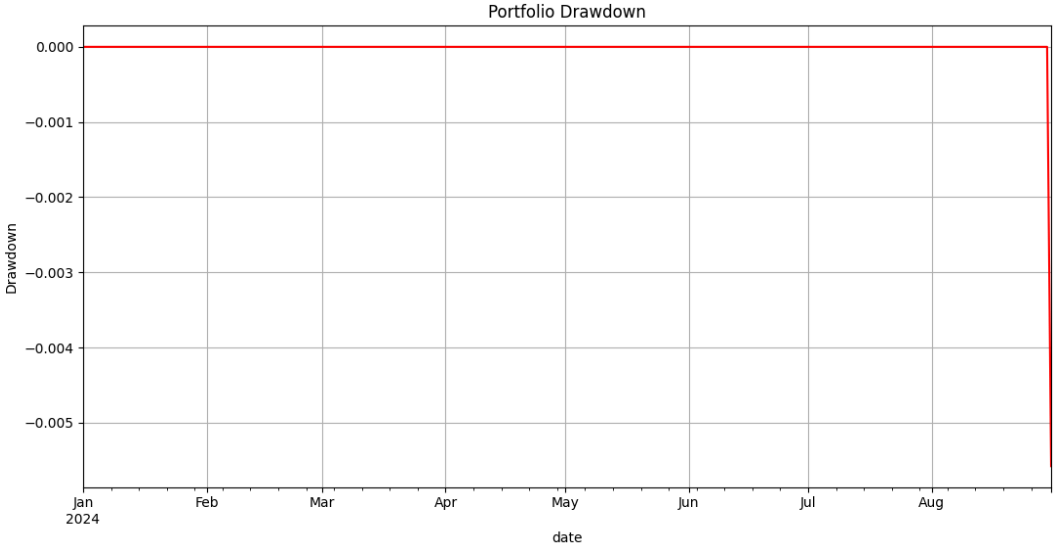
- Maximum drawdown identified: **-0.56%**.
- Computed **Rolling Sharpe** and **Adjusted Rolling Sharpe** ratios for risk-adjusted performance.

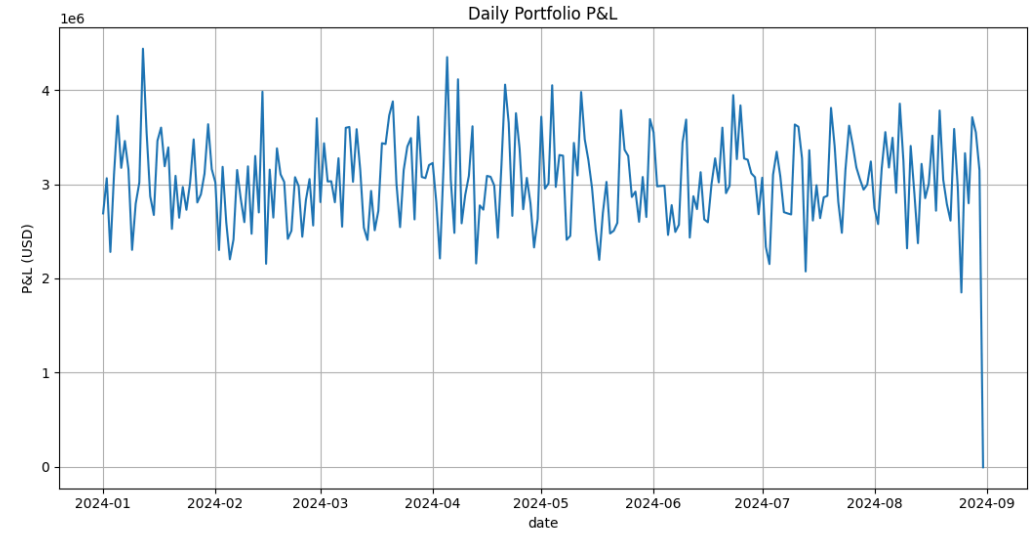
5. Next Steps

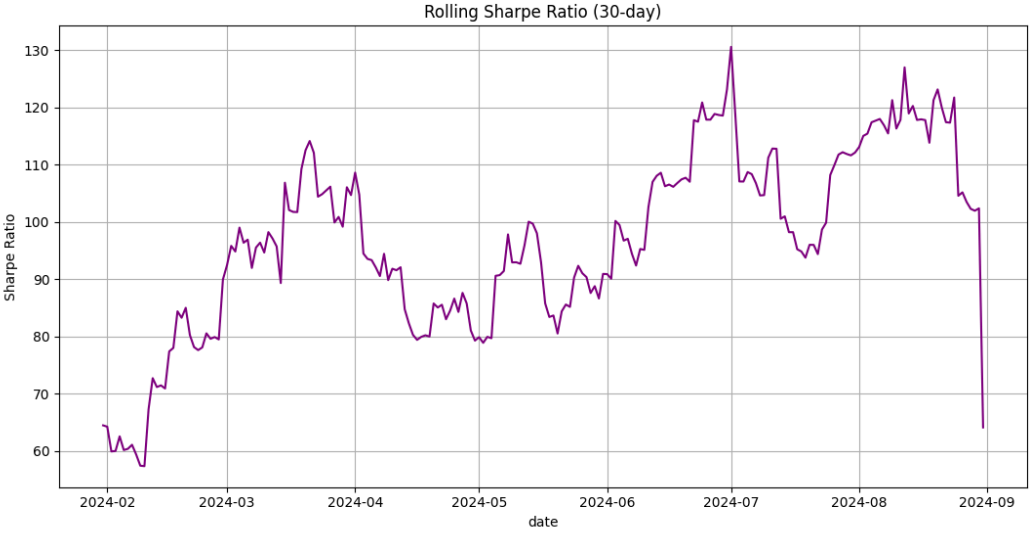
- Add **stress testing** (rate shocks, volatility spikes).
- Automate daily reporting in a clean dashboard format.


📥 Data Loading
(Trading + Instruments)
🔗 Data Integration
(Merge + Snapshots)
📈 Portfolio P&L
(Daily + Strategy-level)
📉 Risk Metrics
(VaR & ES)
📉 Drawdowns
(Max DD)
📊 Sharpe Ratio
(Rolling)
🚀 Next Steps
(Stress Tests + Reports)


Cumulative Portfolio Returns
1e146
4
3
2
1
0
Growth Factor
Jan
2024
Feb
Mar
Apr
May
Jun
Jul
Aug
date


Portfolio Drawdown
0.000
−0.001
−0.002
−0.003
−0.004
−0.005
Drawdown
Jan
2024
Feb
Mar
Apr
May
Jun
Jul
Aug
date


Daily Portfolio P&L
1e6
4
3
2
1
0
P&L (USD)
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07
2024-08
2024-09
date


Rolling Sharpe Ratio (30-day)
130
120
110
100
90
80
70
60
Sharpe Ratio
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07
2024-08
2024-09
date